

Figure 7: Final Decision Matrix

Criterion	Linear Regression	Random Forest	XGBoost	SVM	Selected Model
Predictive Performance	Medium	High	Very High	Medium	Random Forest / XGBoost
Interpretability	High	Medium	Medium-Low	Medium	Linear Regression / Random Forest
Robustness	Medium	High	High	Medium	Random Forest / XGBoost
Computational Cost	Low	Medium	High	Medium	Linear Regression
Deployment Suitability	High	High	Medium	Medium	Random Forest